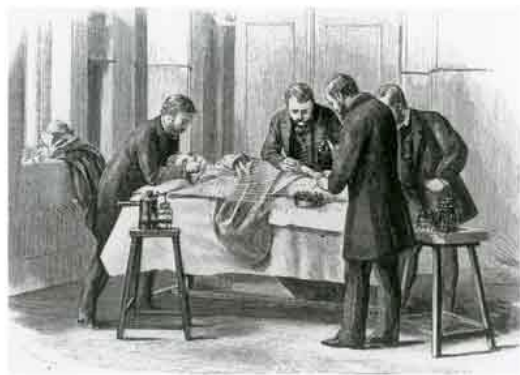




Germ theory

People in the past *didn't understand* how infections occurred. Doctors operated in **dirty, germ-ridden conditions** and thought bad air was to blame. In the mid-19th century, French scientist **LOUIS PASTEUR** showed that some diseases and many infections were caused by bacteria and other microorganisms invading the body.

How it changed the world
Antiseptics helped make surgery cleaner and far safer. Operations became more common and new types of surgery could be developed.

Antiseptics

How microbe-killing substances **CLEANED UP** medicine's act so wounds could heal

Reservoir contains water mixed with carbolic acid.



Pump nozzle sends out a fine mist of carbolic acid.

Handle acts as a lever operating the small pump.

Lister method

British surgeon **JOSEPH LISTER** became convinced that **microbes in the air** were causing infections in wounds, which were usually left open. In the 1860s, Lister started to **clean wounds** and **soak dressings** in carbolic acid—the first antiseptic—which killed many infection-causing microbes. He also built a “**DONKEY ENGINE**” (left) to spray carbolic acid mist throughout his operating theater. Infection and death rates after surgery **plummeted**.



Keeping it clean

From the 1890s onward, surgical instruments were boiled to sterilize them, **eliminating all microbes** before use. Face masks were adopted, and surgeons now **CLEAN THEIR HANDS** with antiseptic solutions before surgery. Rates of infection have been slashed from 50 percent to less than 1 percent.